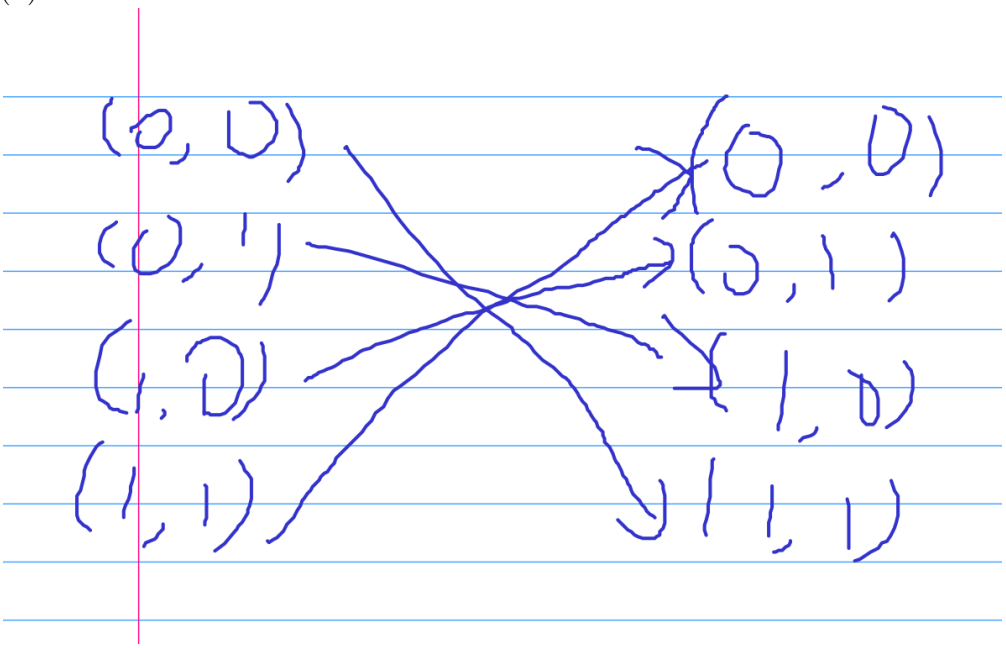
ICS 6B: Homework 5

4.1.4

(a)
Domain = $\{(0,0),(0,1),(1,0),(1,1)\}$

(b)



(c)
Range = $\{(0,0),(0,1),(1,0),(1,1)\}$

4.2.1

(a)
$f(2.2) = \lfloor 2.2 + 1/2 \rfloor = 2$
$f(2.9) = \lfloor 2.9 + 1/2 \rfloor = 3$
$f(2.5) = \lfloor 2.5 + 1/2 \rfloor = 3$
$f(2) = \lfloor 2 + 1/2 \rfloor = 2$
$f(3) = \lfloor 3 + 1/2 \rfloor = 3$

(b)
f(x) rounds x to the nearest integer, and rounds a tie (x ending in .5) up.

(c)
Both round to the nearest integer, but $g(x) = \lfloor x - 1/2 \rfloor$ rounds a tie down instead of up, so f and g differ only when x is a half-integer (f(x) = g(x) + 1 there).

4.2.2

(a)
$f(x) = \lceil \frac{5x}{24} \rceil$
(b)
$g(y) = \lfloor \frac{y}{8} \rfloor$

4.2.4

(b)
Given n is an odd integer. We will show that $\lfloor \frac{n}{2} \rfloor = \frac{n-1}{2}$.

Recall the defining property of the floor: for any real x and any integer m,
$m = \lfloor x \rfloor \iff m \leq x < m + 1$.
So it suffices to find an integer m with $m \leq \frac{n}{2} < m + 1$.

Since n is an odd integer, so $n = 2k + 1$ for some integer k. Then
$\frac{n}{2} = \frac{2k+1}{2} = k + \frac{1}{2}$
and our candidate value is
$\frac{n-1}{2} = \frac{2k}{2} = k$,
which is an integer because k is. Since $\frac{1}{2} > 0$ we have $k < k + \frac{1}{2}$, and since $\frac{1}{2} < 1$ we have $k + \frac{1}{2} < k + 1$. Putting these together,
$k \leq \frac{n}{2} < k + 1$.
Since k is an integer satisfying both bounds, the defining property of the floor gives
$\lfloor \frac{n}{2} \rfloor = k = \frac{n-1}{2}$. ■

4.3.2

(i)
Neither.

Not one-to-one: $f(0,0) = (0,-2) = f(1,0)$ yet $(0,0) \neq (1,0)$.

Not onto: for every $y \in \mathbb{Z}$, $5y - 2 \equiv 3 \pmod 5$, so the second coordinate can never be 0. Thus $(0,0) \in \mathbb{Z} \times \mathbb{Z}$ has no preimage.

(k)
Neither.

Not onto: the smallest value is $f(1,1) = 2 + 1 = 3$, so $1 \in \mathbb{Z}^+$ (and also 2) is never produced.

Not one-to-one: $f(1,3) = 2 + 3 = 5 = 4 + 1 = f(2,1)$ yet $(1,3) \neq (2,1)$.

4.3.6

(b)
True.

Let f be a bijection and $c \neq 0$ a real number. We show that cf is both one-to-one and onto.

One-to-one: suppose $(cf)(x_1) = (cf)(x_2)$ for some $x_1, x_2 \in \mathbb{R}$. Then $c \cdot f(x_1) = c \cdot f(x_2)$. Since $c \neq 0$, we may divide both sides by c, giving $f(x_1) = f(x_2)$. Because f is one-to-one, this implies $x_1 = x_2$. Hence cf is one-to-one.

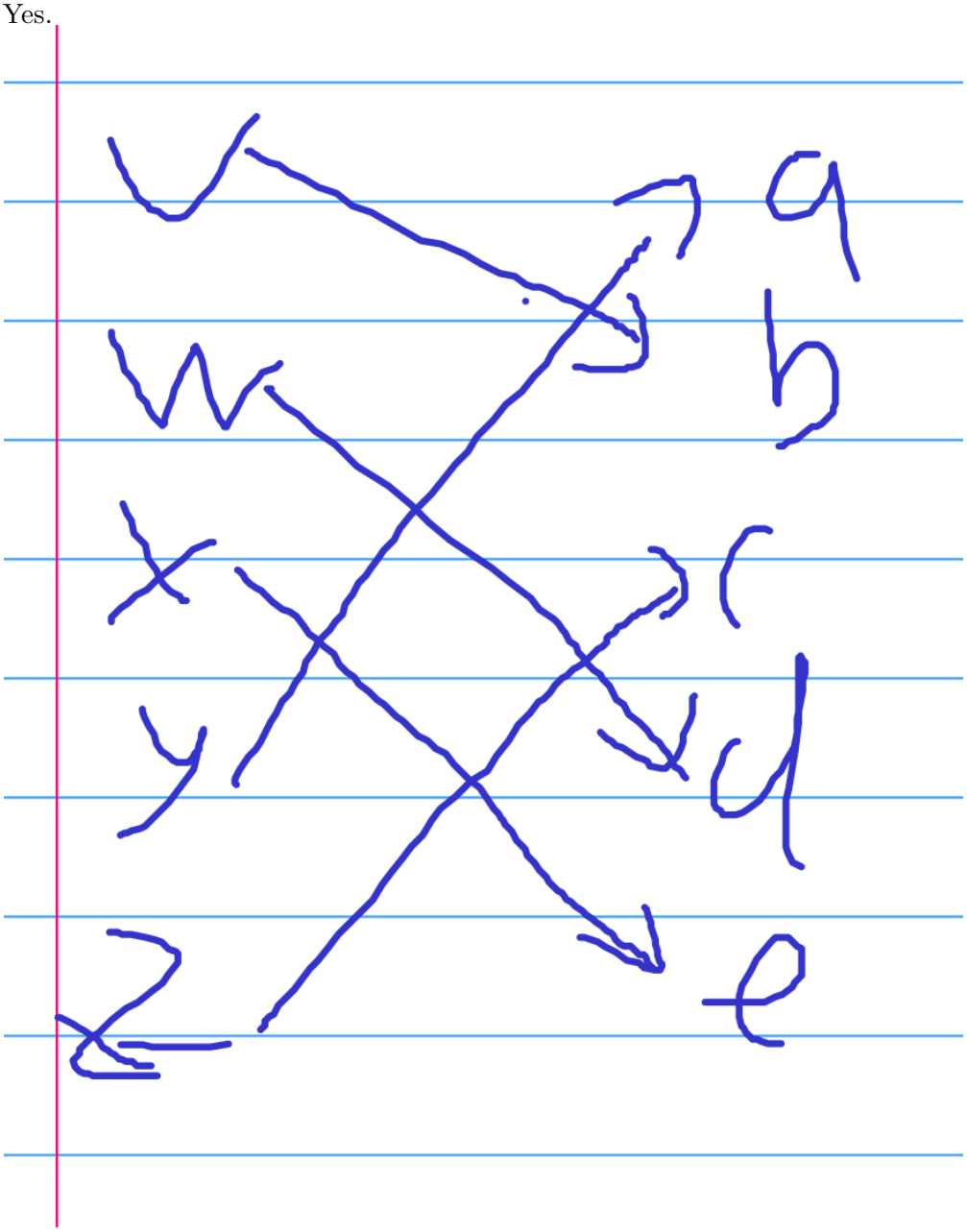
Onto: let $y \in \mathbb{R}$ be arbitrary. Since $c \neq 0$, the number y/c is a real number. Because f is onto, there is some $x \in \mathbb{R}$ with $f(x) = y/c$. Then
$(cf)(x) = c \cdot f(x) = c \cdot \frac{y}{c} = y$.
So every $y \in \mathbb{R}$ has a preimage under cf, meaning cf is onto.

Since cf is both one-to-one and onto, cf is a bijection. ■

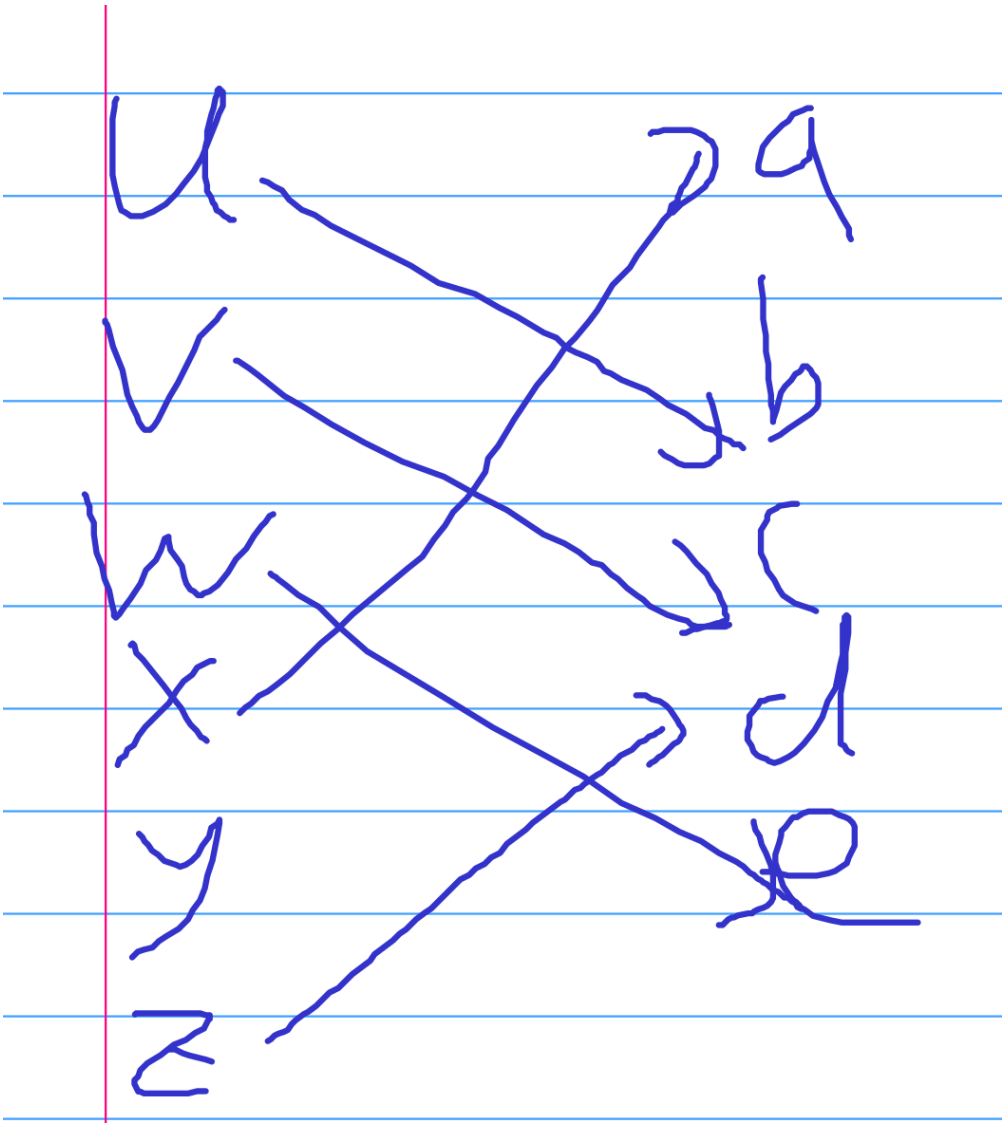
4.4.1

(a)
No.
w has multiple output.

(b)
Yes.



(c)
Yes.


u
v
w
x
y
z
a
b
c
d
e

4.5.2

(b)
$(f \circ h)(52) = f(h(52)) = f(\lceil 52/5 \rceil) = f(11) = 11^2 = 121$

(d)
$(h \circ f)(x) = h(f(x)) = h(x^2) = \lceil \frac{x^2}{5} \rceil$

4.3.4

(d)
One-to-one but not onto.

One-to-one: every input $x_1x_2x_3$ maps to $x_1x_2x_3x_1$, so the first three bits of an output recover its input. If $f(s) = f(t)$, then s and t agree in their first three bits, so $s = t$.

Not onto: in any output the fourth bit equals the first bit. So $0001 \in \{0,1\}^4$ has no preimage, since its preimage would have to be 000, but $f(000) = 0000 \neq 0001$.

(g)
Neither.

Not one-to-one: $f(\{1\}) = \{1\} - \{1\} = \varnothing = \varnothing - \{1\} = f(\varnothing)$ Since, $\{1\} \neq \varnothing$, not one to one.

Not onto: for every $X \subseteq A$, $1 \notin X - B$, so $\{1\} \in P(A)$ has no preimage.

4.4.2

(e)
Yes.

f is its own inverse. For every $X \subseteq A$,
$f(f(X)) = A - (A - X) = X$,
so f is a bijection and
$f^{-1}(X) = A - X$.

(b)
No.

f is not onto: $f(x) = 2x + 34 = 2(x + 17)$ is even for every $x \in \mathbb{Z}$, so $1 \in \mathbb{Z}$ has no preimage. Hence f is not a bijection and has no well-defined inverse.

4.5.6

(b)
$(g \circ h)(010) = g(h(010)) = g(010) = 010$
(h leaves 010 unchanged because its first bit is already the same as its last bit, and reversing 010 gives 010 back.)

(e)
$\{001, 011, 101, 111\}$
The range of f is $\{100, 101, 110, 111\}$, so the range of $g \circ f$ is obtained by reversing each of these. Equivalently, $(g \circ f)(b_1 b_2 b_3) = g(1 b_2 b_3) = b_3 b_2 1$, which is exactly every string whose last bit is 1.